

Accuracy–Interpretability Trade-offs in Exoplanet Transit Modelling, and a Label-Free Physics-Informed Estimator of Planetary Radii

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Transit photometry poses two distinct machine-learning problems: deciding whether a light curve contains a planetary transit, and measuring the planet’s radius ratio R_p/R_* from it. We compare seven model families on both tasks using simulated light curves with exactly known parameters, and score every model on a second axis besides accuracy—the fidelity with which a purely additive surrogate reproduces its output, an explicit and reproducible operationalisation of one dimension of interpretability. The two axes trade off: on detection, the most accurate model (an MLP, $AUC = 0.975$) has the second-lowest additive fidelity ($\mathcal{A} = 0.573$), while the fully readable linear model reaches $AUC = 0.946$ at $\mathcal{A} = 0.999$ —a 0.029 AUC premium for near-total transparency. We then introduce a physics-informed Kolmogorov–Arnold network that predicts transit parameters and is trained *only* to reconstruct the observed flux through a differentiable annular approximation to the Mandel–Agol forward model, never seeing a radius label. It attains $MAE = 0.0100$ in R_p/R_* , within 24% of the best fully supervised model, and occupies a distinct region of the Pareto plane under matched simulation. Applied zero-shot to twelve confirmed Kepler planets it recovers radii with full-sample $MAE = 0.023$, a feasibility result rather than a reliability estimate. We report that full-sample figure as the primary real-data result and show that the label-free reconstruction-residual screen used to identify model mismatch improves it only to 0.018—and rejects one fold whose radius was in fact recovered to within 0.001, demonstrating that reconstruction quality is not a reliable proxy for parameter accuracy.

CCS Concepts: • **Applied computing: Astronomy** • **Computing methodologies: Machine learning** • **Computing methodologies: Neural networks**

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Additional Key Words and Phrases: Exoplanets, interpretable machine learning, Kolmogorov–Arnold networks, physics-informed learning, transit photometry, model comparison.

1 INTRODUCTION

When a planet crosses the disc of its host star along our line of sight, it occults a fraction of the stellar flux and the star appears momentarily fainter. The resulting periodic dip in a light curve—a time series of stellar brightness—is the most productive exoplanet-detection signal known, and it is the mechanism behind the bulk of the confirmed planet catalogue [1, 13].

Machine learning entered this field primarily as a detector. Deep convolutional classifiers now rival or exceed hand-tuned pipelines at separating genuine transits from instrumental and astrophysical false positives [10, 15]. But detection is only half of the scientific product. The dip’s depth encodes the planet’s size, its duration encodes the orbital geometry, and it is those physical parameters, not the detection flag, that enter a population study. A model that classifies well but whose reasoning cannot be inspected is a weaker scientific instrument than one whose inferred quantities can be checked against physics [14].

This paper studies both halves, and studies them along two axes.

1.1 Contributions

- (1) A controlled comparison of seven model families on transit detection and on radius-ratio regression, using a simulator whose ground-truth parameters are known exactly, with a common training and scoring interface for every model.
- (2) An explicit, reproducible interpretability score—additive-surrogate fidelity $\mathcal{A}$ —applied identically to all models, and a Pareto analysis that makes the accuracy–transparency exchange rate quantitative rather than rhetorical.
- (3) A physics-informed KAN that infers R_p/R_* , impact parameter, scaled semi-major axis, and epoch by reconstructing the light curve through a differentiable Mandel–Agol model, using *no* parameter labels at any point in training.
- (4) A zero-shot transfer test on twelve confirmed Kepler planets in which the full sample is the primary reported result, and an explicit demonstration that the residual-based quality screen used to flag model mismatch is not aligned with parameter accuracy.

2 FORWARD MODEL AND SIMULATED DATA

2.1 Transit physics

For a planet of radius ratio $\rho = R_p/R_*$ on a circular orbit of scaled semi-major axis a/R_* and impact parameter b , define phase as $\phi = (t - t_0)/P$, with the simulation period normalised to $P = 1$. The projected star–planet separation is

$$d(\phi) = \sqrt{\left(\frac{a}{R_*}\right)^2 \sin^2(2\pi\phi) + b^2 \cos^2(2\pi\phi)}. \quad (1)$$

The stellar disc is limb darkened: it is brighter at the centre than at the edge. We use the standard quadratic law [2, 5]

$$\frac{I(\mu)}{I(1)} = 1 - u_1(1 - \mu) - u_2(1 - \mu)^2, \quad \mu = \sqrt{1 - r^2}, \quad (2)$$

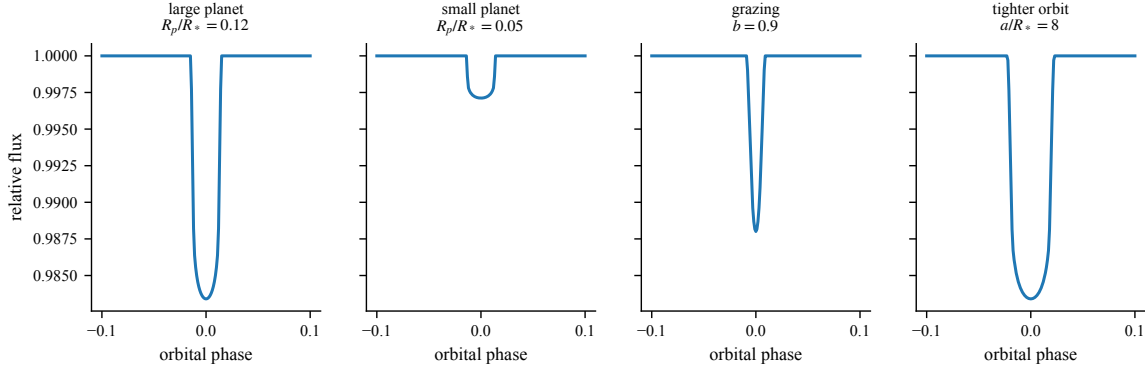


Fig. 1. Transit profiles from Eq. (3). Depth scales approximately as ρ^2 ; a grazing geometry ($b \rightarrow 1$) rounds the profile and destroys the flat bottom; a tighter orbit shortens the crossing. Depth and duration are the two observables from which ρ must be inferred.

with $(u_1, u_2) = (0.3, 0.2)$ throughout. Decomposing the disc into N_r concentric annuli of intensity I_k and computing the occulted area $O(d, R_k, \rho)$ of each analytically [9], the normalised flux is

$$F(\phi) = 1 - \frac{\sum_{k=1}^{N_r} I_k [O(d(\phi), R_k, \rho) - O(d(\phi), R_{k-1}, \rho)]}{\sum_{k=1}^{N_r} I_k \pi (R_k^2 - R_{k-1}^2)}. \quad (3)$$

Equation (3) is differentiable in $(\rho, b, a/R_*, t_0)$, which is what makes the physics-informed estimator of Section 5 trainable. The reference implementation is validated against batman [6]. For 500 parameter draws, its maximum/mean absolute discrepancies and RMSE relative to batman are $5.53 \times 10^{-6}/1.53 \times 10^{-7}/5.50 \times 10^{-7}$. Figure 1 shows how each parameter reshapes the dip.

2.2 Labelled simulation

Each example is a $L = 201$ -point segment spanning orbital phase ± 0.10 . Positives draw $\rho \sim \mathcal{U}(0.03, 0.15)$, $b \sim \mathcal{U}(0, 0.95)$, $a/R_* \sim \mathcal{U}(8, 18)$, and a small epoch jitter, then add Gaussian noise of per-point scale $\sigma = 10^u$, $u \sim \mathcal{U}(-3.3, -2.0)$. Negatives are noise-only; half additionally carry a low-frequency sinusoid imitating intrinsic stellar variability, which forces the detector to learn the transit *shape* rather than merely “something changed”. The transit signal-to-noise ratio is

$$\text{SNR} = \frac{\delta}{\sigma/\sqrt{n_{\text{in}}}}, \quad \delta = 1 - \min_{\phi} F(\phi), \quad (4)$$

with n_{in} the number of in-transit samples implied by the analytic duration. This is a simulation-specific white-noise proxy, not a claim about survey detection completeness under correlated noise or systematics. The training set holds 8,000 segments (4,006 positive), the disjoint test set 3,000 (1,483 positive).

Preprocessing differs by task, and the difference is substantive. For detection each curve is standardised,

$$\tilde{x}_i = \frac{x_i - \text{median}(x)}{\text{std}(x)}, \quad (5)$$

which discards absolute depth—appropriate, because depth is a nuisance for the presence/absence question. For characterisation, depth *is* the signal, so we subtract the median without rescaling.

3 MODELS AND THE INTERPRETABILITY AXIS

3.1 The seven families

All models consume the same 201-point vector and expose a common `fit/score` interface: regularised linear (logistic for detection, ridge for regression), random forest, gradient-boosted trees, a three-layer MLP, a 1-D CNN with two convolution–pooling blocks, a bidirectional LSTM, and a Kolmogorov–Arnold network (KAN) [8].

The KAN merits a note because it is the reason interpretability is measurable rather than assumed. A conventional network applies fixed nonlinearities to learned linear combinations; a KAN instead learns a spline ψ_j per input coordinate and sums them,

$$g(\mathbf{x}) = \sum_{j=1}^L \psi_j(x_j), \quad (6)$$

which is a generalised additive model [4] with learned shape functions. This describes the single-layer [201, 1] KAN (5,632 parameters; cubic splines, grid 5). The physics-KAN is a two-layer [201, 8, 4] model (28,796 parameters), not a GAM; it is assessed by surrogate fidelity. Only the single-layer KAN’s ψ_j are directly plotted without post-hoc approximation [12].

The MLP/CNN/LSTM have 34,177/3,233/9,025 parameters and train for 30/30/18 epochs at 10^{-3} ; the single-layer and physics KANs train for 40 at 5×10^{-3} and 80 at 3×10^{-3} (batch 256). Forest/XGBoost use 200 trees (XGBoost depth 4), and ridge $\alpha = 1$. Settings were fixed a priori without tuning or compute matching. The physics-KAN also evaluates 128 annuli per batch; wall-clock times were not logged.

3.2 Additive-surrogate fidelity

Interpretability has no canonical scalar. We therefore measure one specific, well-defined property: how closely the model behaves like a sum of per-coordinate functions over an independent probe distribution. Given a model output $s(\mathbf{x})$ —mapped to the logit scale for classifiers—we fit a cubic-spline additive surrogate

$$\hat{s}(\mathbf{x}) = \beta_0 + \sum_{j=1}^L h_j(x_j), \quad h_j \in \text{span}\{B\text{-splines}\}, \quad (7)$$

by ridge regression on two thirds of a fresh 3,000-curve probe set, and define

$$\mathcal{A} = R^2(s, \hat{s}) \text{ on the held-out third.} \quad (8)$$

$\mathcal{A} = 1$ means the model is exactly additive over the probe distribution; $\mathcal{A} \approx 0$ means its behaviour is dominated by interactions no additive explanation can capture.

We are explicit about what (8) does and does not measure. It is reproducible, applied identically to every model, and is a genuine property of the learned function. It does *not* certify that an explanation is causally correct, stable under distribution shift, physically meaningful, or useful to a human. It measures one dimension of interpretability, and the sanity check that the linear model scores $\mathcal{A} = 0.999$ confirms the measurement behaves as intended.

Table 1. Detection on the held-out synthetic set. $\mathcal{A}$ is additive-surrogate fidelity, Eq. (8). “Readable” marks models whose learned function can be inspected directly without a surrogate.

Model	Family	ROC-AUC	$\mathcal{A}$	Readable
MLP	neural net	0.975	0.573	no
XGBoost	tree	0.964	0.822	no
Random forest	tree	0.959	0.903	no
CNN	neural net	0.956	0.768	no
Logistic/Ridge	linear	0.946	0.999	yes
KAN	KAN	0.943	0.900	yes
LSTM	neural net	0.906	0.728	no

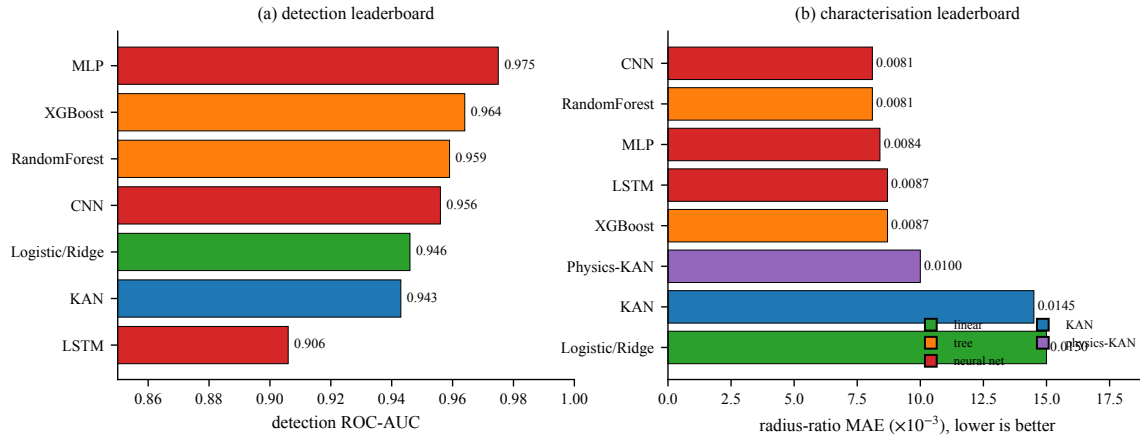


Fig. 2. (a) Detection ROC-AUC and (b) characterisation MAE in R_p/R_* , coloured by model family. Note the axis scales: the detection spread is 0.069 AUC across all seven families, whereas the characterisation spread is nearly a factor of two in error.

4 TASK 1: DETECTION

Table 1 and Figure 2(a) give held-out ROC-AUC together with $\mathcal{A}$. Every family exceeds AUC = 0.90; the problem is not hard in aggregate, and the differences live at low SNR.

Two results are worth isolating. First, the LSTM is the weakest model (AUC = 0.906) despite being the family nominally designed for sequences. A transit is a localised, translation-tolerant shape; convolution and even a fixed linear template match that structure better than a recurrence that must carry information across 201 steps. Second, the KAN reaches AUC = 0.943, within 0.003 of the linear model, while retaining direct readability. Because Eq. (6) sums learned per-bin functions, the model’s sensitivity profile across orbital phase can be plotted and compared against the true mean transit template; in the source study that comparison yields a correlation of 0.94, i.e. the KAN independently rediscovered the transit shape rather than latching onto an artefact.

As a limited optimisation-variance check, we retrained the leading MLP with five seeds on the same split. Test AUC is 0.9751 ± 0.0009 (sample SD), with range [0.9736, 0.9757]; its 0.029 gap from the linear model is therefore much larger than the observed MLP seed variation. Other families and the physics-KAN were not replicated, so their pairwise differences remain single-seed results.

Table 2. Characterisation of R_p/R_* on the held-out synthetic set. Skill is $1 - \text{MAE}/\Delta R_p$ with $\Delta R_p = 0.12$ the sampled range. The physics-KAN is the only entry trained without radius labels.

Model	Family	MAE	Skill	$\mathcal{A}$	Labels used
Random forest	tree	0.0081	0.933	0.476	yes
CNN	neural net	0.0081	0.933	0.426	yes
MLP	neural net	0.0084	0.930	0.298	yes
XGBoost	tree	0.0087	0.928	0.314	yes
LSTM	neural net	0.0087	0.928	0.484	yes
Physics-KAN	physics-KAN	0.0100	0.917	0.530	no
KAN	KAN	0.0145	0.879	0.989	yes
Logistic/Ridge	linear	0.0150	0.875	0.997	yes

5 TASK 2: CHARACTERISATION

5.1 A label-free physics-informed estimator

Supervised regression of ρ requires labelled radii, which for real systems come from published fits and are therefore scarce and heterogeneous. We remove that dependence. A KAN maps the light curve to four bounded physical parameters via a sigmoid reparameterisation,

$$\theta_k = \ell_k + (u_k - \ell_k) \sigma(z_k(\mathbf{x})), \quad \theta = (\rho, b, a/R_*, t_0), \quad (9)$$

those parameters are pushed through the differentiable forward model (3), and the network is trained on the reconstruction objective alone:

$$\mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_{i=1}^N \frac{1}{L} \sum_{j=1}^L \left(F_{\theta(\mathbf{x}_i)}(\phi_j) - 1 - o_{ij} \right)^2, \quad (10)$$

where o_{ij} is the median-subtracted observed flux. No radius label appears anywhere in (10). Radius strongly controls depth, but it is not uniquely identifiable from depth: ρ , b , a/R_* , limb darkening, and t_0 jointly shape depth, duration, and ingress. The bounded joint fit exploits all 201 samples, while remaining subject to these degeneracies—the same inverse-problem logic that underlies physics-informed neural networks [11].

A finite-difference Jacobian check quantifies this coupling: range-scaled median RMS flux sensitivities over 200 curves are 0.00618 (ρ), 0.00261 (b), 0.00303 (a/R_*), and 0.00819 (t_0). With geometry known, fitting ρ while assuming $(u_1, u_2) = (0.3, 0.2)$ gives MAE 0.00194 for true (0.1, 0.1) and 0.00240 for true (0.5, 0.3) over 120 curves (maxima 0.00855/0.01095). Unknown geometry can add error, so these are diagnostics, not robustness guarantees.

5.2 Results

Table 2 and Figure 2(b) give the leaderboard. The best supervised model (random forest) reaches MAE = 0.0081. The physics-informed KAN reaches 0.0100—24% worse, while using *zero* radius labels—and beats both label-supervised readable models, the KAN (0.0145) and ridge (0.0150), by a wide margin.

The physics-informed model’s advantage over the readable supervised models is structural. Ridge and the plain KAN must both express ρ as an additive function of per-bin flux, but ρ enters the forward model through Eq. (3) nonlinearly and jointly with b and a/R_* : depth alone does not determine radius when the geometry is unknown. The physics-KAN does not have to learn that coupling from data, because the coupling is built into its decoder.

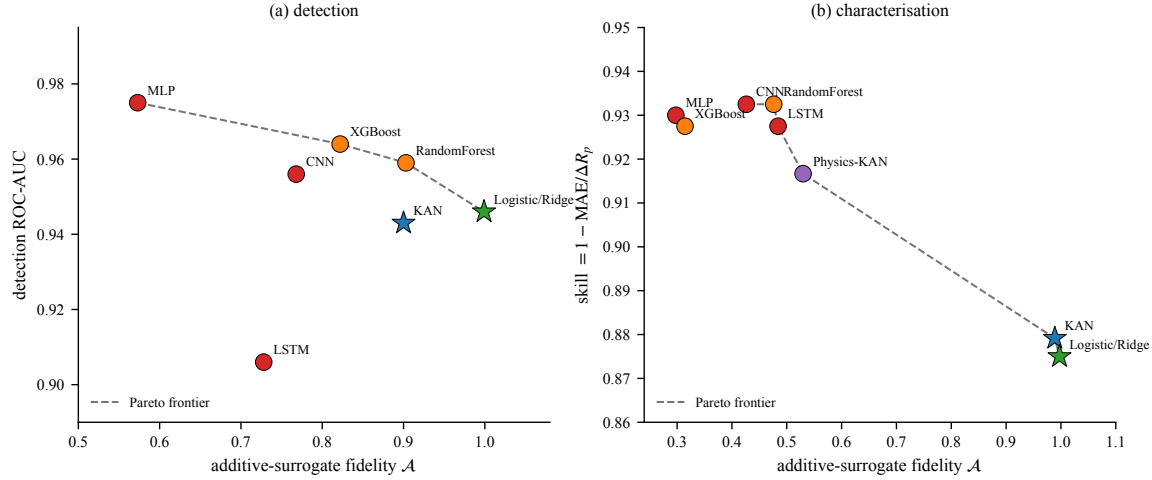


Fig. 3. Accuracy against additive-surrogate fidelity. Stars mark directly readable models, circles black boxes; the dashed line is the Pareto frontier. On detection (a) the frontier is short and the cost of transparency is 0.029 AUC. On characterisation (b) the frontier is long: full readability costs 85% higher error than the best black box, while the physics-informed model occupies an intermediate position.

5.3 The Pareto view

Figure 3 places every model on the accuracy–additivity plane. Three observations follow.

First, the trade-off is real but its exchange rate is task dependent. On detection, moving from the MLP ($\mathcal{A} = 0.573$) to the fully readable linear model ($\mathcal{A} = 0.999$) costs 0.029 AUC—cheap. On characterisation, the same move costs 85% higher error ($0.0081 \rightarrow 0.0150$)—expensive.

Second, high $\mathcal{A}$ does not imply high accuracy and low $\mathcal{A}$ does not imply the reverse. The MLP has both the best detection AUC and nearly the worst additivity, but the random forest achieves $\mathcal{A} = 0.903$ at AUC = 0.959—nearly linear-model transparency at nearly MLP accuracy. Assuming a strict frontier between the two axes is not warranted by these data.

Third, the physics-informed model is not on the Pareto frontier of Figure 3(b) and this is the correct result: on *these* axes it is dominated. Its distinguishing property is orthogonal to both—it requires no labels—which is precisely the situation where a two-axis Pareto plot is an incomplete decision aid.

6 ZERO-SHOT TRANSFER TO KEPLER

The physics-informed model was trained exclusively on simulation and has never seen a real light curve. We apply it, without adaptation, to twelve confirmed Kepler planets whose long-cadence photometry has been folded on the published orbital period and binned to the same 201-point grid [7].

Table 3. Zero-shot recovery of R_p/R_* on twelve confirmed Kepler planets. “Screen” is the label-free reconstruction-residual rule: ✓ retained, × rejected as suspected model mismatch. The screen rejects Kepler-1624 b despite an error of 0.001.

Planet	Published	Recovered	$ \Delta $	Screen
Kepler-719 b	0.080	0.081	0.001	✓
Kepler-1624 b	0.109	0.108	0.001	×
Kepler-855 b	0.074	0.081	0.007	✓
Kepler-550 b	0.049	0.060	0.011	✓
Kepler-546 b	0.056	0.069	0.013	✓
Kepler-228 d	0.040	0.055	0.015	✓
Kepler-734 b	0.042	0.060	0.018	✓
Kepler-170 b	0.030	0.052	0.022	✓
Kepler-244 b	0.031	0.054	0.023	✓
Kepler-485 b	0.121	0.076	0.045	×
Kepler-548 b	0.122	0.070	0.052	✓
Kepler-12 b	0.119	0.046	0.073	×
Full-sample MAE (primary)			0.023	
Screened-subset MAE (exploratory, $n = 9$)			0.018	

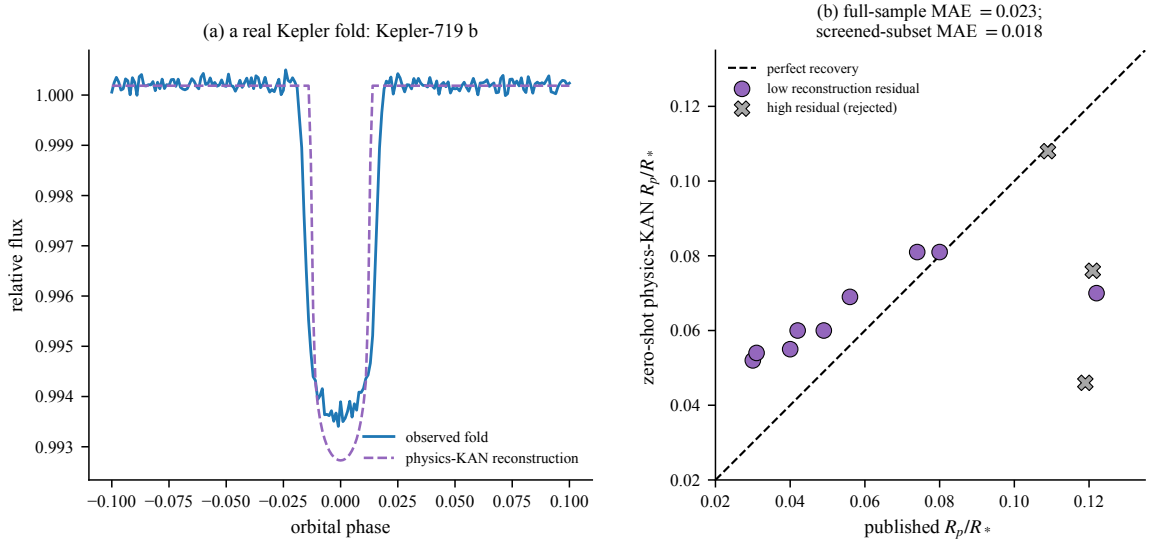


Fig. 4. (a) An observed Kepler fold with the physics-KAN’s reconstruction overlaid. (b) Recovered against published radius ratio for all twelve planets; markers indicate the label-free reconstruction-residual screen. The three largest published radii are all underestimated, the signature of long-cadence smearing.

6.1 Reading the result honestly

The full twelve-planet MAE is 0.023, and that is the primary number. It is roughly $2.3\times$ the model’s synthetic error, which is the expected penalty for sim-to-real transfer: real folds carry detrending residuals, contamination from neighbouring stars, and—most importantly—30-minute long-cadence integration, which smears short ingress and egress and systematically shallows the profile.

That smearing explains the dominant error pattern in Figure 4(b). The three largest published radii (Kepler-485 b, Kepler-548 b, Kepler-12 b, all $\rho > 0.12$) are underestimated by 0.045–0.073, whereas the small-radius planets are mildly overestimated. Deep transits of short duration lose the most to integration, so this is a forward-model mismatch, not random error.

6.2 The residual screen does not track accuracy

The source study screens folds by reconstruction residual, a label-free diagnostic for model mismatch. Replicating that rule exactly, retaining the lowest-residual 75% improves MAE from 0.023 to 0.018—a modest gain, not the dramatic improvement a threshold chosen after seeing the data might suggest.

More informative is which folds the screen selects. It correctly rejects Kepler-12 b (error 0.073) and Kepler-485 b (error 0.045), but it also rejects Kepler-1624 b, whose radius was recovered to within 0.001—the joint-best result in the sample—while retaining Kepler-548 b, whose error of 0.052 is the second worst. Reconstruction residual measures how well the assumed forward model explains the fold, which is not the same quantity as how accurately ρ was recovered; a fold can be noisy yet yield a good depth estimate, or clean yet be fit with a compensating combination of b and a/R_* . Any confirmatory study should fix the quality threshold in advance and report accepted and rejected cases separately, as we have done here.

7 DISCUSSION

7.1 Supported findings

- (1) Accuracy and additive-surrogate fidelity are distinct objectives with a task-dependent exchange rate: 0.029 AUC for detection, 85% relative error for characterisation.
- (2) A differentiable transit model permits training a radius estimator by reconstruction alone, within 24% of fully supervised accuracy and far ahead of supervised readable models.
- (3) Zero-shot transfer to twelve real Kepler folds is feasible (MAE = 0.023) but is not evidence of population-level reliability; its dominant error is underestimation of deep transits under long-cadence smearing.
- (4) Reconstruction residual is a poor proxy for parameter accuracy and should not be used as a post-hoc accuracy filter.

7.2 Threats to validity

Training and test simulations share a generator, parameter ranges, and noise model, so synthetic scores measure interpolation within a known family rather than robustness to unmodelled systematics. The physics-informed model enjoys a matched-model advantage: its decoder is the very process that generated the synthetic data. The limb-darkening sensitivity check and Kepler results show how accuracy degrades when that match is broken, but do not span realistic mismatch. Additive-surrogate fidelity captures one dimension of interpretability and is measured on a probe distribution drawn from the same simulator. Twelve planets is a demonstration, not a population estimate, and the sample is biased towards well-characterised systems. Only the MLP detection result is repeated across five seeds; all other training results carry unquantified optimisation variance, so close pairwise differences should not be treated as meaningful.

7.3 Future work

The next experiment should test the current model under correlated noise, limb-darkening mismatch, and long-cadence integration; adding the latter to Eq. (3) should address the deep-transit bias in Figure 4. TESS [13] and single-transit data [3] would then test broader transfer.

8 CONCLUSION

On simulated transits, full transparency costs 0.029 detection AUC and 85% more radius error. A physics-informed alternative reaches near-supervised accuracy without parameter labels under matched simulation. Its twelve-planet transfer is a feasibility check that exposes instrument mismatch, not a validated catalogue-level estimator.

REPRODUCIBILITY

Code, notebooks, and fixed inputs are in the public repository. The figure script regenerates the figures, tables, and residual screen; the notebook validates Eq. (3) against batman [6] and fixes all data-generation seeds. Committed outputs used NumPy 2.5.1, pandas 2.3.3, Matplotlib 3.11.0, scikit-learn 1.9.0, XGBoost 2.1.4, PyTorch 2.6.0+cu124, batman 2.5.3, and pykan 0.2.8.

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